

Isaac Diaz

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Education

California State Polytechnic University, Pomona *College of Science*

Bachelor of Science, Computer Science

May 2023

Mt. San Antonio College

Associate Degree in Science, Mathematics

February 2021

Relevant Skills

- C++
- Java
- Dart Programming Language
- Unreal Engine 5 (C++/Blueprints)
- Unity/C#
- MySQL

Academic Projects

Ray Tracer Project, *Computer Graphics*, Fall 2022

- Used Java Processing to create a ray tracer that is capable of rendering spheres, planes, triangles, and various quadrics by using trigonometry, linear algebra, and vector math.
- Implemented graphics systems such as the Phong Lighting Model, reflections and refractions, and Complex Solid Geometry features that are applied to primitive shapes.

California Wildfire Prediction, *Software Engineering*, Fall 2022

- Simulated an Agile Software Development Life Cycle where I acted as Project Manager to create a machine learning model through Amazon Web Services to predict wildfires in California.
- Used AWS to create an ML model through preprocessed/cleaned wildfire data in California. Created AWS Lambda functions that was capable of receiving user data and returning the probability of a wildfire in several counties in California to be displayed on a web app made in AWS.

Pomona Transit System, *Software Engineering*, Spring 2023

- Used Java, JDBC (Java Database Connectivity), and MySQL to create a transit system database that is capable of displaying and modifying bus stop information present in the database.
- The system could display and edit the schedule of all trips for a given starting location, display the stops of a given trip, add a driver, add or delete a driven bus, and insert data of given trip offering.

Restaurant Point of Sales System, *Software Engineering*, Spring 2022

- Used Java, JDBC (Java Database Connectivity), Java Swing and MySQL to create a restaurant Point of Sale system that could replace a physical receipt-based restaurant with a digital-based GUI system.
- The system can create receipts of orders of various food items for specific tables in a representative layout of a restaurant. It can create, update, add, and delete new food items, orders, receipts, and tables that would all be updated in the database.

C++ Graph Data Structure, *Data Structures*, Spring 2020

- Implemented the graph data structure using strong knowledge of C++.
- Used advanced C++ features such as pointers, templates, decorators, operator overloading, and object-oriented programming to implement Dijkstra's Algorithm and model an airport flight system.
- I've used C++ to implement many data structures and projects for my college career.

Club, *Cal Poly Pomona Game Development Club*, 2022 - 2023

- Collaborated with a team of people to complete tasks and create a minimum viable product for games.